
PrimeCore Minerals

Local Area Network (LAN) Configuration

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Introduction

A correctly configured switch is important in a company's network infrastructure as it can act as another barrier in the company's security implementation by assisting in network security, supply the company with a reliable network plus manage network traffic to prohibit the whole network being flooded with unwanted and unnecessary traffic. Understanding the role of a switch compared to other network equipment as well understanding that these switches are to manage traffic on the 2nd layer of the OSI model¹ plus the ability to replicate the switch configuration and view the switch configuration is critical in achieving this goal and as each switch manufacturer has their own switch configuration and command line interface (CLI),

To obtain these goals, a correctly configured switch with the correct features is required as well as the ability to duplicate its configuration to maintain network stability.

This has been achieved in this virtual environment however it is understood that it will need to be translated from OVS CLI commands to Cisco CLI commands without missing any of the features configured when this layout is implemented on the Cisco switches.

Configuration

To replicate the network LAN environment, Open vSwitch (OVS) has been installed on top of a Linux Server 26.04 image and is the main controller for all switch configurations. The features of a Cisco switch that have been replicated and are configured on these switches are as follows:

- Unique hostname for each switch
- Timezone set to Queensland (location of main office)
- Securing of network configuration files within Linux Server
- Creating and managing virtual networks (VLANs)
- Creating and managing trunk ports
- Turning on Rapid Spanning Tree Protocol
- Turning on LLDP (equivalent to Cisco cdp protocol)
- Setting port maximum speeds
- Enabling a banner (warning label displayed to user before they log in)
- Testing of bandwidth speeds

Some other applications were required to enable these features and they are

- net-tools
- lldpd
- iperf3

LAN Switch Modifications to Physical Device

The port speeds on the switches have been limited to 100Mbps for the standard ports with an uplink speed of 1Gbps for the trunk ports due to the demonstration being in a virtual environment and there are limitations on the underlying software and hardware like BUS speed, RAM used, disk speed, etc that can affect the results of our tests if they are placed at the higher end of the speed limits that the physical switch is actually capable of.

The trunk ports will be done through the SPF+ ports in the module so the company can utilize a 10Gbps connection. To represent this in the virtual environment, the trunk ports have been allocated a 1Gbps speed limit, irrespective of which port they are connected to. Most of the trunk ports are the last 3 ports where possible.

The physical setup has a total of 32 ports however in our environment, the switches have been configured with only 12 ports due to hardware and resource limitation in our virtual environment.

Due to this limitation of port speed and number of ports assigned to each switch in the virtual environment, control-plane intelligence hasn't been enabled on the switch as the other limitations will prohibit the switch from getting near the physical switches' limitation.

Switch Template

To obtain consistency between all LAN switches, a LAN switch template was created and is called the PCM_LAN_Switch_Template.pdf. It is a separate document and is part of the technical artifact documentation.

Switch port speed

To confirm the speed limit between devices, iperf3 has been installed on the switches so any switches can be linked up to test the speed, reliability and jitter between the switches.

To test the switches, log into the receiving switch and run:

- Sudo iperf -s

The sending switch runs

- Sudo iperf3 -c <Receiving Switch IP Address> -t 10

Switch port speed will be displayed under each switch section

Switch Configuration Reporting

As there are multiple applications installed to obtain the same features as a Cisco switch, there is no standard command to view the port and switch configuration so a script has been written and saved on the switches and APs so that it can be run anytime in order to view the switch and port status. The script is part of the artefact documents and is called PCM_Report_Script.pdf

The script scans the switch configuration and allows a quick view of the switch configuration. The script set up is:

Save file at /usr/local/bin/show-port-status

Open file sudo nano /usr/local/bin/show-port-status

To make file runnable run: sudo chmod +x /usr/local/bin/show-port-status

Transfer script to file show-port-status > /var/log/port-status-<SWITCH NAME>.txt

To view the file: sudo nano /var/log/port-status.txt

The switch report for each switch is below with the following features

- Each section shows in blue and has ### before and after it
- Name of switch/AP
- The timezone the switch/AP is set to
 - o Important for diagnosing switch log times
- Overview of switch features and if they are enabled/disabled including:
 - o Multicasting snooping (prevents broadcasting flooding)
 - o Name of bridge containing all ports (Open vSwitch feature)
 - o RSTP (enabled is PCM configuration requirement)
 - o STP (disabled by PCM configuration requirement as its legacy compared to RSTP)
 - o LLDP/CDP (used to confirm if a port is connected to another switch/AP)
- Port/Bond-ring configuration
 - o Name (port name/number. Table for each switch below translates between physical port number, EVE-NG Port Name and Switch config name.
 - o VLAN Tag – if only 1 VLAN on port, it shows here. If it's a trunk port, it will be blank. This will match the 'Port VLAN (tag) OR TRUNK type & number (using OVS names)' column below for each switch
 - o Trunks – if port is a trunk (and blank in the VLAN Tag name) then it will list the VLANs that travel over that port.

- o Port Speed (QoS) – QoS has been enabled on each port to replicate the physical speed limit that ports have
- o Port Description – a description of what device is connected to that port. Useful for network engineers to troubleshoot
- o LLDP Neighbour – if port is connected to another switch/AP then it will be discovered by LLDP and script shows the result here.
- o Port MAC – used to confirm the MAC that the port is using. Required in virtual environment to identify if MAC have been manually assigned or still default as default MAC address has high chance of being replicated on another switch and causing network inconsistency.

Switch configuration setup and reports

The following tables and screenshots show the table setup and the report output for each switch.

PCM-SITEA-SW01 switch report

PCM-SITEA-SW01 IP address: 192.168.199.10

Table 1 – port configuration on switch PCM-SITEA-SW01

Port Number	EVE-NG Port Name	Switch config name	Port VLAN (tag) OR TRUNK type & number (using OVS names)	Port connected to:	MAC Address	Switch speed
1	E0	ens3	tag=1	Spare port	26:13:23:6a:10:01	100Mbps
2	E1	ens4	tag=41	CCTV Camera	26:13:23:6a:10:02	100Mbps
3	E2	ens5	tag=90	CCTV Server	26:13:23:6a:10:03	100Mbps
4	E3	ens6	tag=90	Samba Server	26:13:23:6a:10:04	100Mbps
5	E4	ens7	tag=1	Spare port	26:13:23:6a:10:05	100Mbps
6	E5	ens8	tag=90	Printer Server	26:13:23:6a:10:06	100Mbps
7	E6	ens9	trunks=10,11,20,30,40,41,50,90,91,99	WAN (pfSense firewall/router)	26:13:23:6a:10:07	1Gbps

8	E7	ens10	tag=1	Spare port	26:13:2 3:6a:10: 08	100M bps
9	E8	ens11	trunks=11,20,99	PCM-SITEA- AP01	26:13:2 3:6a:10: 09	100M bps
10	E9	ens12	bond-ring (trunks=10,11,20,30,40,41,50 ,90,91,99)	PCM-SITED- SW01	26:13:2 3:6a:10: 10	1Gbps
11	E10	ens13	bond-ring (trunks=10,11,20,30,40,41,50 ,90,91,99)	PCM-SITEA- SW02	26:13:2 3:6a:10: 11	1Gbps
12	E11	ens14	tag=99	pfSense Management PC	26:13:2 3:6a:10: 12	1Gbps
vlan9 9	vlan99	vlan99		vlan99	26:13:2 3:6a:99: 10	NA

```

GNU nano 8.7.1 /var/log/port-status-PCM-SITEA-SW01.txt
### SWITCH NAME ###
PCM-SITEA-SW01

### TIMEZONE ###
Time zone: Australia/Brisbane (AEST, +1000)

### BRIDGE (RSTP, LLDP) ###
mcast_snooping_enable: true
name : br0
rstp_enable : true
stp_enable : false
LLDP/CDP Status: LLDP Enabled, CDP Enabled

### PORTS (VLANs, trunks, speed limits, LLDP) ###

### bond-ring ###
Name : bond-ring
VLAN Tag : []
Trunks : [10, 11, 20, 30, 40, 41, 50, 90, 91, 99]
Port Speed (QoS) : 1Gbps
Port Description : stack link to PCM-SITEA-SW02
LLDP Neighbour : ens13 -> PCM-SITEA-SW02
LLDP Neighbour : ens12 -> PCM-SITED-SW01
Port MAC : bond-ring link

### ens3 ###
Name : ens3
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:10:01

### ens4 ###
Name : ens4
VLAN Tag : 41
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : CCTV_Camera
LLDP Neighbour : None
Port MAC : 26:13:23:6a:10:02

```

Figure 1 – PCM-SITEA-SW01 – Report output screenshot 1 of 3

```

GNU nano 8.7.1 /var/log/port-status-PCM-SITEA-SW01.txt
### ens5 ###
Name : ens5
VLAN Tag : 90
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : CCTV_Server
LLDP Neighbour : None
Port MAC : 26:13:23:6a:10:03

### ens6 ###
Name : ens6
VLAN Tag : 90
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : Samba(AD)_Server
LLDP Neighbour : None
Port MAC : 26:13:23:6a:10:04

### ens7 ###
Name : ens7
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:10:05

### ens8 ###
Name : ens8
VLAN Tag : 90
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : Printer_Server
LLDP Neighbour : None
Port MAC : 26:13:23:6a:10:06

### ens9 ###
Name : ens9
VLAN Tag : []
Trunks : [10, 11, 20, 30, 40, 41, 50, 90, 91, 99]
Port Speed (QoS) : 1Gbps
Port Description : WAN_Link
LLDP Neighbour : None
Port MAC : 26:13:23:6a:10:07

```

Figure 2 – PCM-SITEA-SW01 – Report output screenshot 2 of 3

```
GNU nano 8.7.1 /var/log/port-status-PCM-SITEA-SW01.txt
### ens9 ###
Name : ens9
VLAN Tag : []
Trunks : [10, 11, 20, 30, 40, 41, 50, 90, 91, 99]
Port Speed (QoS) : 1Gbps
Port Description : WAN_Link
LLDP Neighbour : None
Port MAC : 26:13:23:6a:10:07

### ens10 ###
Name : ens10
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:10:08

### ens11 ###
Name : ens11
VLAN Tag : []
Trunks : [11, 20, 99]
Port Speed (QoS) : 1Gbps
Port Description : connected to PCM-SITEA-AP01
LLDP Neighbour : PCM-SITEA-AP01
Port MAC : 26:13:23:6a:10:09

### ens14 ###
Name : ens14
VLAN Tag : 99
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : pfSense Managment PC
LLDP Neighbour : None
Port MAC : 26:13:23:6a:10:12

### vlan99 ###
Name : vlan99
VLAN Tag : 99
Trunks : []
Port Speed (QoS) : Unlimited
Port Description : None
LLDP Neighbour : None
Port MAC : 26:13:23:6a:99:10
```

Figure 3 – PCM-SITEA-SW01 – Report output screenshot 3 of 3

PCM-SITEA-SW02 switch report

PCM-SITEA-SW02 IP address: 192.168.199.11

Table 2 – port configuration on switch PCM-SITEA-SW02

Port Number	EVE-NG Port Name	Switch config name	Port VLAN (tag) OR TRUNK type & number (using OVS names)	Port connected to:	MAC Address	Switch speed
1	E0	ens3	tag=10	LAN Employee PC	26:13:23:6a:11:01	100Mbps
2	E1	ens4	tag=91	Backup Server	26:13:23:6a:11:02	100Mbps
3	E2	ens5	tag=91	Archive Server	26:13:23:6a:11:03	100Mbps
4	E3	ens6	tag=40	VC unit	26:13:23:6a:11:04	100Mbps
5	E4	ens7	tag=30	Printer	26:13:23:6a:11:05	100Mbps
6	E5	ens8	tag=90	Web Server	26:13:23:6a:11:06	100Mbps
7	E6	ens9	tag=1	Spare port	26:13:23:6a:11:07	100Mbps
8	E7	ens10	tag=1	Spare port	26:13:23:6a:11:08	100Mbps
9	E8	ens11	tag=1	Spare port	26:13:23:6a:11:09	100Mbps
10	E9	ens12	trunk=11,20,99	PCM-SITEA-AP02	26:13:23:6a:11:10	1Gbps
11	E10	ens13	Bond-ring (trunk=10,11,20,30,40,41,50,90,91,99)	Bond-ring	26:13:23:6a:11:11	1Gbps
12	E11	ens14	Bond-ring (trunk=10,11,20,30,40,41,50,90,91,99)	Bond-ring	26:13:23:6a:11:12	1Gbps
vlan99	vlan99	vlan99		vlan99	26:13:23:6a:99:11	NA

```

GNU nano 8.7.1 /var/log/port-status-PCM-SITEA-SW02.txt
### SWITCH NAME ###
PCM-SITEA-SW02

### TIMEZONE ###
Time zone: Australia/Brisbane (AEST, +1000)

### BRIDGE (RSTP, LLDP) ###
mcast_snooping_enable: true
name : br0
rstp_enable : true
stp_enable : false
LLDP/CDP Status: LLDP Enabled, CDP Enabled

### PORTS (VLANs, trunks, speed limits, LLDP) ###

### bond-ring ###
Name : bond-ring
VLAN Tag : []
Trunks : [10, 11, 20, 30, 40, 41, 50, 90, 91, 99]
Port Speed (QoS) : 1Gbps
Port Description : stack link to PCM-SITEA-SW01
LLDP Neighbour : ens14 -> PCM-SITEA-SW01
LLDP Neighbour : ens13 -> PCM-SITEB-SW01
Port MAC : bond-ring link

### ens3 ###
Name : ens3
VLAN Tag : 10
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : LAN_Employee
LLDP Neighbour : None
Port MAC : 26:13:23:6a:11:01

### ens4 ###
Name : ens4
VLAN Tag : 91
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : Backup_Server
LLDP Neighbour : None
Port MAC : 26:13:23:6a:11:02

```

Figure 4 – PCM-SITEA-SW02 – Report output screenshot 1 of 3

```
GNU nano 8.7.1 /var/log/port-status-PCM-SITEA-SW02.txt
### ens5 ###
Name : ens5
VLAN Tag : 91
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : Archive_Server
LLDP Neighbour : None
Port MAC : 26:13:23:6a:11:03

### ens6 ###
Name : ens6
VLAN Tag : 40
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : VC_Unit
LLDP Neighbour : None
Port MAC : 26:13:23:6a:11:04

### ens7 ###
Name : ens7
VLAN Tag : 30
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : Printer
LLDP Neighbour : None
Port MAC : 26:13:23:6a:11:05

### ens8 ###
Name : ens8
VLAN Tag : 90
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:11:06

### ens9 ###
Name : ens9
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:11:07
```

Figure 5 – PCM-SITEA-SW02 – Report output screenshot 2 of 3

```
GNU nano 8.7.1 /var/log/port-status-PCM-SITEA-SW02.txt
### ens9 ###
Name : ens9
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:11:07

### ens10 ###
Name : ens10
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:11:08

### ens11 ###
Name : ens11
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:11:09

### ens12 ###
Name : ens12
VLAN Tag : []
Trunks : [11, 20, 99]
Port Speed (QoS) : 1Gbps
Port Description : connected to PCM-SITEA-AP02
LLDP Neighbour : PCM-SITEA-AP02
Port MAC : 26:13:23:6a:11:10

### vlan99 ###
Name : vlan99
VLAN Tag : 99
Trunks : []
Port Speed (QoS) : Unlimited
Port Description : None
LLDP Neighbour : None
Port MAC : 26:13:23:6a:99:11
```

Figure 6 – PCM-SITEA-SW02 – Report output screenshot 3 of 3

PCM-SITEB-SW01 switch report

PCM-SITEB-SW01 IP address: 192.168.199.20

Table 3 – port configuration on switch PCM-SITEB-SW01

Port Number	EVE-NG Port Name	Switch config name	Port VLAN (tag) OR TRUNK type & number (using OVS names)	Port connected to:	MAC Address	Switch speed
1	E0	ens3	tag=10	LAN Employee PC	26:13:23:6a:20:01	100Mbps
2	E1	ens4	tag=40	VC unit	26:13:23:6a:20:02	100Mbps
3	E2	ens5	tag=30	Printer	26:13:23:6a:20:03	100Mbps
4	E3	ens6	tag=50	IoT device	26:13:23:6a:20:04	100Mbps
5	E4	ens7	tag=41	CCTV camera	26:13:23:6a:20:05	100Mbps
6	E5	ens8	tag=1	Spare port	26:13:23:6a:20:06	100Mbps
7	E6	ens9	tag=1	Spare port	26:13:23:6a:20:07	100Mbps
8	E7	ens10	tag=1	Spare port	26:13:23:6a:20:08	100Mbps
9	E8	ens11	tag=1	Spare port	26:13:23:6a:20:09	100Mbps
10	E9	ens12	trunk=11,20,99	PCM-SITEB-AP01	26:13:23:6a:20:10	1Gbps
11	E10	ens13	trunk=10,11,20,30,40,41,50,90,91,99	PCM-SITEC-SW01	26:13:23:6a:20:11	1Gbps
12	E11	ens14	trunk=10,11,20,30,40,41,50,90,91,99	PCM-SITEA-SW02	26:13:23:6a:20:12	1Gbps
vlan99	vlan99	vlan99	NA	vlan99	26:13:23:6b:99:10	NA

```

GNU nano 8.7.1 /var/log/port-status-PCM-SITEB-SW01.txt
### SWITCH NAME ###
PCM-SITEB-SW01

### TIMEZONE ###
Time zone: Australia/Brisbane (AEST, +1000)

### BRIDGE (RSTP, LLDP) ###
mcast_snooping_enable: true
name : br0
rstp_enable : true
stp_enable : false
LLDP/CDP Status: LLDP Enabled, CDP Enabled

### PORTS (VLANs, trunks, speed limits, LLDP) ###

### ens3 ###
Name : ens3
VLAN Tag : 10
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : LAN_Employee
LLDP Neighbour : None
Port MAC : 26:13:23:6a:20:01

### ens4 ###
Name : ens4
VLAN Tag : 40
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : VC_Unit
LLDP Neighbour : None
Port MAC : 26:13:23:6a:20:02

### ens5 ###
Name : ens5
VLAN Tag : 30
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : Printer
LLDP Neighbour : None
Port MAC : 26:13:23:6a:20:03

```

Figure 7 – PCM-SITEB-SW01 – Report output screenshot 1 of 3

```
GNU nano 8.7.1 /var/log/port-status-PCM-SITEB-SW01.txt
### ens6 ###
Name : ens6
VLAN Tag : 50
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : IoT_Device
LLDP Neighbour : None
Port MAC : 26:13:23:6a:20:04

### ens7 ###
Name : ens7
VLAN Tag : 41
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : CCTV_Camera
LLDP Neighbour : None
Port MAC : 26:13:23:6a:20:05

### ens8 ###
Name : ens8
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:20:06

### ens9 ###
Name : ens9
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:20:07

### ens10 ###
Name : ens10
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:20:08
```

Figure 8 – PCM-SITEB-SW01 – Report output screenshot 2 of 3

```
GNU nano 8.7.1 /var/log/port-status-PCM-SITEB-SW01.txt
### ens11 ###
Name : ens11
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:20:09

### ens12 ###
Name : ens12
VLAN Tag : []
Trunks : [11, 20, 99]
Port Speed (QoS) : 1Gbps
Port Description : connected to PCM-SITEB-AP01
LLDP Neighbour : None
Port MAC : 26:13:23:6a:20:10

### ens13 ###
Name : ens13
VLAN Tag : []
Trunks : [10, 11, 20, 30, 40, 41, 50, 90, 91, 99]
Port Speed (QoS) : 1Gbps
Port Description : connected to PCM-SITEC-SW01
LLDP Neighbour : PCM-SITEC-SW01
Port MAC : 26:13:23:6a:20:11

### ens14 ###
Name : ens14
VLAN Tag : []
Trunks : [10, 11, 20, 30, 40, 41, 50, 90, 91, 99]
Port Speed (QoS) : 1Gbps
Port Description : connected to PCM-SITEA-SW02
LLDP Neighbour : PCM-SITEA-SW02
Port MAC : 26:13:23:6a:20:12

### vlan99 ###
Name : vlan99
VLAN Tag : 99
Trunks : []
Port Speed (QoS) : Unlimited
Port Description : None
LLDP Neighbour : None
Port MAC : 26:13:23:6a:99:20
```

Figure 9 – PCM-SITEB-SW01 – Report output screenshot 3 of 3

PCM-SITEC-SW01 switch report

PCM-SITEC-SW01 IP address: 192.168.199.30

Table 4 – port configuration on switch PCM-SITEC-SW01

Port Number	EVE-NG Port Name	Switch config name	Port VLAN (tag) OR TRUNK type & number (using OVS names)	Port connected to:	MAC Address	Switch speed
1	E0	ens3	tag=10	LAN Employee PC	26:13:23:6a:30:01	100Mbps
2	E1	ens4	tag=40	VC unit	26:13:23:6a:30:02	100Mbps
3	E2	ens5	tag=30	Printer	26:13:23:6a:30:03	100Mbps
4	E3	ens6	tag=50	IoT device	26:13:23:6a:30:04	100Mbps
5	E4	ens7	tag=41	CCTV camera	26:13:23:6a:30:05	100Mbps
6	E5	ens8	tag=1	Spare port	26:13:23:6a:30:06	100Mbps
7	E6	ens9	tag=1	Spare port	26:13:23:6a:30:07	100Mbps
8	E7	ens10	tag=1	Spare port	26:13:23:6a:30:08	100Mbps
9	E8	ens11	tag=1	Spare port	26:13:23:6a:30:09	100Mbps
10	E9	ens12	trunk=11,20,99	PCM-SITEC-AP01	26:13:23:6a:30:10	1Gbps
11	E10	ens13	trunk=10,11,20,30,40,41,50,90,91,99	PCM-SITEC-SW01	26:13:23:6a:30:11	1Gbps
12	E11	ens14	trunk=10,11,20,30,40,41,50,90,91,99	PCM-SITEA-SW02	26:13:23:6a:30:12	1Gbps
vlan99	vlan99	vlan99	NA	vlan99	26:13:23:6b:99:10	NA

```

GNU nano 8.7.1 /var/log/port-status-PCM-SITEC-SW01.txt
### SWITCH NAME ###
PCM-SITEC-SW01

### TIMEZONE ###
Time zone: Australia/Brisbane (AEST, +1000)

### BRIDGE (RSTP, LLDP) ###
mcast_snooping_enable: true
name : br0
rstp_enable : true
stp_enable : false
LLDP/CDP Status: LLDP Enabled, CDP Enabled

### PORTS (VLANs, trunks, speed limits, LLDP) ###

### ens3 ###
Name : ens3
VLAN Tag : 10
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : LAN_Employee
LLDP Neighbour : None
Port MAC : 26:13:23:6a:30:01

### ens4 ###
Name : ens4
VLAN Tag : 40
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : VC_Unit
LLDP Neighbour : None
Port MAC : 26:13:23:6a:30:02

### ens5 ###
Name : ens5
VLAN Tag : 30
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : Printer
LLDP Neighbour : None
Port MAC : 26:13:23:6a:30:03

```

Figure 10 – PCM-SITEC-SW01 – Report output screenshot 1 of 3

```
GNU nano 8.7.1 /var/log/port-status-PCM-SITEC-SW01.txt
### ens6 ###
Name : ens6
VLAN Tag : 50
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : IoT_Device
LLDP Neighbour : None
Port MAC : 26:13:23:6a:30:04

### ens7 ###
Name : ens7
VLAN Tag : 41
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : CCTV_Camera
LLDP Neighbour : None
Port MAC : 26:13:23:6a:30:05

### ens8 ###
Name : ens8
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:30:06

### ens9 ###
Name : ens9
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:30:07

### ens10 ###
Name : ens10
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:30:08
```

Figure 11 – PCM-SITEC-SW01 – Report output screenshot 2 of 3

```
GNU nano 8.7.1 /var/log/port-status-PCM-SITEC-SW01.txt
### ens11 ###
Name : ens11
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:30:09

### ens12 ###
Name : ens12
VLAN Tag : []
Trunks : [11, 20, 99]
Port Speed (QoS) : 1Gbps
Port Description : connected to PCM-SITEC-AP01
LLDP Neighbour : None
Port MAC : 26:13:23:6a:30:10

### ens13 ###
Name : ens13
VLAN Tag : []
Trunks : [10, 11, 20, 30, 40, 41, 50, 90, 91, 99]
Port Speed (QoS) : 1Gbps
Port Description : connected to PCM-SITED-SW01
LLDP Neighbour : PCM-SITED-SW01
Port MAC : 26:13:23:6a:30:11

### ens14 ###
Name : ens14
VLAN Tag : []
Trunks : [10, 11, 20, 30, 40, 41, 50, 90, 91, 99]
Port Speed (QoS) : 1Gbps
Port Description : connected to PCM-SITEB-SW02
LLDP Neighbour : PCM-SITEB-SW01
Port MAC : 26:13:23:6a:30:12

### vlan99 ###
Name : vlan99
VLAN Tag : 99
Trunks : []
Port Speed (QoS) : Unlimited
Port Description : None
LLDP Neighbour : None
Port MAC : 26:13:23:6a:90:30
```

Figure 12 – PCM-SITEC-SW01 – Report output screenshot 3 of 3

PCM-SITED-SW01 switch report

PCM-SITED-SW01 IP address: 192.168.199.40

Table 5 – port configuration on switch PCM-SITED-SW01

Port Number	EVE-NG Port Name	Switch config name	Port VLAN (tag) OR TRUNK type & number (using OVS names)	Port connected to:	26:13:23:6a:40:01	Switch speed
1	E0	ens3	tag=10	LAN Employee PC	26:13:23:6a:40:02	100Mbps
2	E1	ens4	tag=40	VC unit	26:13:23:6a:40:03	100Mbps
3	E2	ens5	tag=30	Printer	26:13:23:6a:40:04	100Mbps
4	E3	ens6	tag=50	IoT device	26:13:23:6a:40:05	100Mbps
5	E4	ens7	tag=41	CCTV camera	26:13:23:6a:40:06	100Mbps
6	E5	ens8			26:13:23:6a:40:07	100Mbps
7	E6	ens9			26:13:23:6a:40:08	100Mbps
8	E7	ens10			26:13:23:6a:40:09	100Mbps
9	E8	ens11			26:13:23:6a:40:10	100Mbps
10	E9	ens12	trunk=11,20,99	PCM-SITED-AP01	26:13:23:6a:40:11	1Gbps
11	E10	ens13	trunk=10,11,20,30,40,41,50,90,91,99	PCM-SITEC-SW01	26:13:23:6a:40:12	1Gbps
12	E11	ens14	trunk=10,11,20,30,40,41,50,90,91,99	PCM-SITEA-SW01	26:13:23:6a:40:13	1Gbps
vlan99	vlan99	vlan99	NA	vlan99	26:13:23:6b:99:10	NA

```

GNU nano 8.7.1 /var/log/port-status-pcm-sited-sw01.txt
### SWITCH NAME ###
PCM-SITED-SW01

### TIMEZONE ###
Time zone: Australia/Brisbane (AEST, +1000)

### BRIDGE (RSTP, LLDP) ###
mcast_snooping_enable: true
name : br0
rstp_enable : true
stp_enable : false
LLDP/CDP Status: LLDP Enabled, CDP Enabled

### PORTS (VLANs, trunks, speed limits, LLDP) ###

### ens3 ###
Name : ens3
VLAN Tag : 10
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : LAN_Employee
LLDP Neighbour : None
Port MAC : 26:13:23:6a:40:01

### ens4 ###
Name : ens4
VLAN Tag : 40
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : VC_Unit
LLDP Neighbour : None
Port MAC : 26:13:23:6a:40:02

### ens5 ###
Name : ens5
VLAN Tag : 30
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : Printer
LLDP Neighbour : None
Port MAC : 26:13:23:6a:40:03

```

Figure 13 – PCM-SITED-SW01 – Report output screenshot 3 of 3

```

GNU nano 8.7.1 /var/log/port-status-pcm-sited-sw01.txt
### ens6 ###
Name : ens6
VLAN Tag : 50
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : IoT_Device
LLDP Neighbour : None
Port MAC : 26:13:23:6a:40:04

### ens7 ###
Name : ens7
VLAN Tag : 41
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : CCTV_Camera
LLDP Neighbour : None
Port MAC : 26:13:23:6a:40:05

### ens8 ###
Name : ens8
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:40:06

### ens9 ###
Name : ens9
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:40:07

### ens10 ###
Name : ens10
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:40:08

```

Figure 14 – PCM-SITED-SW01 – Report output screenshot 2 of 3

```
GNU nano 8.7.1 /var/log/port-status-pcm-sited-sw01.txt
### ens11 ###
Name : ens11
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:40:09

### ens12 ###
Name : ens12
VLAN Tag : []
Trunks : [11, 20, 99]
Port Speed (QoS) : 1Gbps
Port Description : connected to PCM-SITED-AP01
LLDP Neighbour : None
Port MAC : 26:13:23:6a:40:10

### ens13 ###
Name : ens13
VLAN Tag : []
Trunks : [10, 11, 20, 30, 40, 41, 50, 90, 91, 99]
Port Speed (QoS) : 1Gbps
Port Description : connected to PCM-SITEC-SW01
LLDP Neighbour : PCM-SITEC-SW01
Port MAC : 26:13:23:6a:40:11

### ens14 ###
Name : ens14
VLAN Tag : []
Trunks : [10, 11, 20, 30, 40, 41, 50, 90, 91, 99]
Port Speed (QoS) : 1Gbps
Port Description : connected to PCM-SITEA-SW01
LLDP Neighbour : PCM-SITEA-SW01
Port MAC : 26:13:23:6a:40:12

### vlan99 ###
Name : vlan99
VLAN Tag : 99
Trunks : []
Port Speed (QoS) : Unlimited
Port Description : None
LLDP Neighbour : None
Port MAC : 26:13:23:6a:99:40
```

Figure 15 – PCM-SITED-SW01 – Report output screenshot 3 of 3

PCM-SITEA-AP01 switch report

PCM-SITEA-AP01 IP address: 192.168.199.110

Table 6 – port configuration on switch PCM-SITEA-AP01

Port Number	EVE-NG Port Name	Switch config name	Port VLAN (tag) OR TRUNK type & number (using OVS names)	Port connected to:	MAC Address	Switch speed
1	E0	ens3	tag=11	Employee_WiFi	26:13:23:6a:60:01	100Mbps
2	E1	ens4	tag=20	Visitor_WiFi	26:13:23:6a:60:02	100Mbps
3	E2	ens5	tag=1	spare port	26:13:23:6a:60:03	100Mbps
4	E3	ens6	tag=1	spare port	26:13:23:6a:60:04	100Mbps
5	E4	ens7	tag=1	spare port	26:13:23:6a:60:05	100Mbps
6	E5	ens8	tag=1	spare port	26:13:23:6a:60:06	100Mbps
7	E6	ens9	tag=1	spare port	26:13:23:6a:60:07	100Mbps
8	E7	ens10	trunks=11,20,99	PCM-SITEA-SW01	26:13:23:6a:60:08	1Gbps

```

GNU nano 8.7.1 /var/log/port-status-pcm-sitea-ap01.txt
### SWITCH NAME ###
PCM-SITEA-AP01

### TIMEZONE ###
Time zone: Australia/Brisbane (AEST, +1000)

### BRIDGE (RSTP, LLDP) ###
mcast_snooping_enable: true
name : br0
rstp_enable : true
stp_enable : false
LLDP/CDP Status: LLDP Enabled, CDP Enabled

### PORTS (VLANs, trunks, speed limits, LLDP) ###

### ens3 ###
Name : ens3
VLAN Tag : 11
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : WiFi_Employee
LLDP Neighbour : None
Port MAC : 26:13:23:6a:60:01

### ens4 ###
Name : ens4
VLAN Tag : 20
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : WiFi_Visitor
LLDP Neighbour : None
Port MAC : 26:13:23:6a:60:02

```

Figure 16 – PCM-SITEA-AP01 – Report output screenshot 1 of 3

```
GNU nano 8.7.1 /var/log/port-status-pcm-sitea-ap01.txt
### ens5 ###
Name : ens5
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:60:03

### ens6 ###
Name : ens6
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:60:04

### ens7 ###
Name : ens7
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:60:05

### ens8 ###
Name : ens8
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:60:06
```

Figure 17 – PCM-SITEA-AP01 – Report output screenshot 2 of 3

```
GNU nano 8.7.1 /var/log/port-status-pcm-sitea-ap01.txt
### ens7 ###
Name : ens7
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:60:05

### ens8 ###
Name : ens8
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:60:06

### ens9 ###
Name : ens9
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:60:07

### ens10 ###
Name : ens10
VLAN Tag : []
Trunks : [11, 20, 99]
Port Speed (QoS) : 1Gbps
Port Description : connected to PCM-SITEA-SW01
LLDP Neighbour : PCM-SITEA-SW01
Port MAC : 26:13:23:6a:60:08

### vlan99 ###
Name : vlan99
VLAN Tag : 99
Trunks : []
Port Speed (QoS) : Unlimited
Port Description : None
LLDP Neighbour : None
Port MAC : 26:13:23:6a:99:60
```

Figure 18 – PCM-SITEA-AP01 – Report output screenshot 3 of 3

PCM-SITEA-AP02 switch report

PCM-SITEA-AP02 IP address: 192.168.199.111

Table 7 – port configuration on switch PCM-SITEA-AP01

Port Number	EVE-NG Port Name	Switch config name	Port VLAN (tag) OR TRUNK type & number (using OVS names)	Port connected to:	MAC Address	Switch speed
1	E0	ens3	tag=11	Employee_WiFi	26:13:23:6a:61:01	100Mbps
2	E1	ens4	tag=20	Visitor_WiFi	26:13:23:6a:61:02	100Mbps
3	E2	ens5	tag=1	spare port	26:13:23:6a:61:03	100Mbps
4	E3	ens6	tag=1	spare port	26:13:23:6a:61:04	100Mbps
5	E4	ens7	tag=1	spare port	26:13:23:6a:61:05	100Mbps
6	E5	ens8	tag=1	spare port	26:13:23:6a:61:06	100Mbps
7	E6	ens9	tag=1	spare port	26:13:23:6a:61:07	100Mbps
8	E7	ens10	trunks=11,20,99	PCM-SITEA-SW01	26:13:23:6a:61:08	1Gbps

```

GNU nano 8.7.1 /var/log/port-satus-pcm-sitea-ap02.txt
### SWITCH NAME ###
PCM-SITEA-AP02

### TIMEZONE ###
Time zone: Australia/Brisbane (AEST, +1000)

### BRIDGE (RSTP, LLDP) ###
mcast_snooping_enable: true
name : br0
rstp_enable : true
stp_enable : false
LLDP/CDP Status: LLDP Enabled, CDP Enabled

### PORTS (VLANs, trunks, speed limits, LLDP) ###

### ens3 ###
Name : ens3
VLAN Tag : 11
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : WiFi_Employee
LLDP Neighbour : None
Port MAC : 26:13:23:6a:61:01

### ens4 ###
Name : ens4
VLAN Tag : 20
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : Visitor_Employee
LLDP Neighbour : None
Port MAC : 26:13:23:6a:61:02

```

Figure 19 – PCM-SITEA-AP02 – Report output screenshot 1 of 3

```

GNU nano 8.7.1 /var/log/port-satus-pcm-sitea-ap02.txt
### ens5 ###
Name : ens5
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:61:03

### ens6 ###
Name : ens6
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:61:04

### ens7 ###
Name : ens7
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:61:05

### ens8 ###
Name : ens8
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:61:06

```

Figure 20 – PCM-SITEA-AP02 – Report output screenshot 2 of 3

```

GNU nano 8.7.1 /var/log/port-satus-pcm-sitea-ap02.txt
### ens7 ###
Name : ens7
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:61:05

### ens8 ###
Name : ens8
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:61:06

### ens9 ###
Name : ens9
VLAN Tag : 1
Trunks : []
Port Speed (QoS) : 100Mbps
Port Description : spare port
LLDP Neighbour : None
Port MAC : 26:13:23:6a:61:07

### ens10 ###
Name : ens10
VLAN Tag : []
Trunks : [11, 20, 99]
Port Speed (QoS) : 1Gbps
Port Description : connected to PCM-SITEA-SW02
LLDP Neighbour : PCM-SITEA-SW02
Port MAC : 26:13:23:6a:61:08

### vlan99 ###
Name : vlan99
VLAN Tag : 99
Trunks : []
Port Speed (QoS) : Unlimited
Port Description : None
LLDP Neighbour : None
Port MAC : 26:13:23:6a:99:61

```

Figure 21 – PCM-SITEA-AP02 – Report output screenshot 3 of 3

Reference

1. Forouzan, BA 2021, *Data communications and networking with TCP/IP protocol suite*, 6th edition, McGraw-Hill.